

Question ID: 42f19012

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which expression is equivalent to $a^{\frac{11}{12}}$, where $a > 0$?

Answer

- A. $\sqrt[12]{a^{132}}$
- B. $\sqrt[144]{a^{132}}$
- C. $\sqrt[121]{a^{132}}$
- D. $\sqrt[11]{a^{132}}$

Correct Answer: B

Rationale

Choice B is correct. Since $\frac{12}{12} = 1$, multiplying the exponent of the given expression by $\frac{12}{12}$ yields an equivalent expression: $a^{(\frac{11}{12})(\frac{12}{12})} = a^{(\frac{132}{144})}$. Since $\frac{132}{144} = 132(\frac{1}{144})$, the expression $a^{\frac{132}{144}}$ can be rewritten as $a^{(132)(\frac{1}{144})}$. Applying properties of exponents, this expression can be rewritten as $(a^{132})^{\frac{1}{144}}$. An expression of the form $(m)^{\frac{1}{k}}$, where $m > 0$ and $k > 0$, is equivalent to $\sqrt[k]{m}$. Therefore, $(a^{132})^{\frac{1}{144}}$ is equivalent to $\sqrt[144]{a^{132}}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.